

A Supervised Learning Surrogate for Joint Waveform and Altitude Optimization

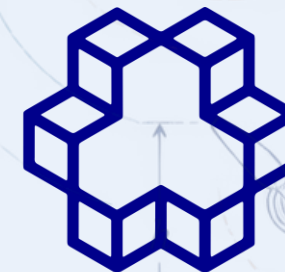
UAV-enabled integrated sensing and wireless power transfer

Machine Learning Course Project

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UAV

OPTIMIZATION



MACHINE LEARNING



One airborne platform must serve two competing tasks

The same transmit waveform supports radar sensing and wireless power transfer.

Therefore, the EHDS can harvest energy from the radar probing signals [20,21].



Figure 1. Illustration of UAV-enabled integrated RadWPT system.

- Radar sensing favors a desired beampattern around target directions.
- Wireless power transfer favors energy delivery toward ground devices.
- UAV altitude changes both path loss and array geometry, so the two decisions are coupled.

Design question

Can previous optimized scenarios predict a useful altitude without discarding waveform constraints?

Source: Kang et al. (2024), Fig. 1

The objective balances radar and power fidelity

$$\text{P1} \quad \min \quad \rho \cdot L_{\text{rad}}(X, \gamma) + (1 - \rho) \cdot L_{\text{wpt}}(X, h)$$

decision variables

waveform covariance X · scaling γ · UAV altitude h

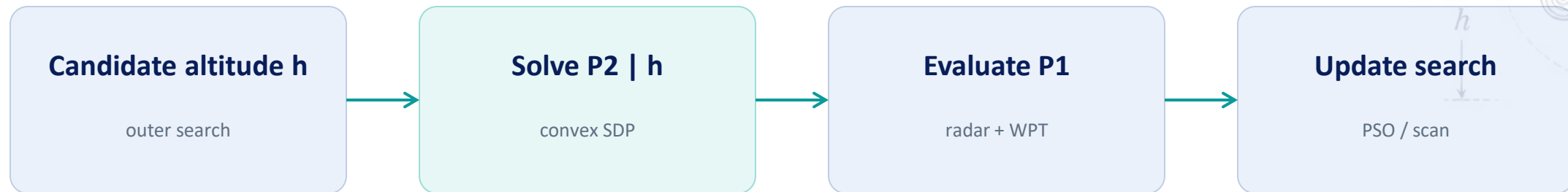
physical constraints

- $X \succeq 0$ — positive semidefinite covariance
- $\text{diag}(X) = P_T / N$ — fixed per-antenna power
- $H_{\min} \leq h \leq H_{\max}$

Why P1 is difficult

For a fixed altitude, $P2|h$ is convex. Across altitude, the propagation-dependent objective is nonlinear and non-convex.

Kang searches altitude outside the exact SDP



Algorithm 1

Accurate joint design
Exact P2 at every outer evaluation

73 exact P2 solves

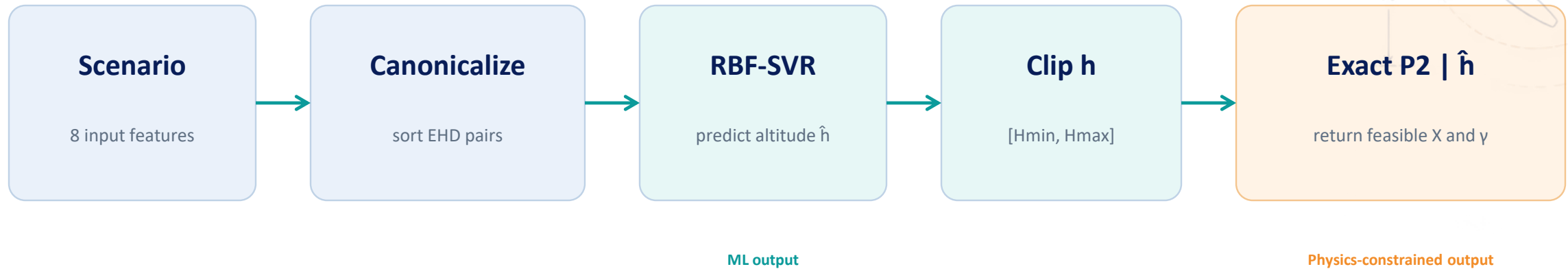
Algorithm 2

Restricted waveform structure
Lower complexity, near-optimal result

$\approx 0.15\text{--}3.08\%$ gap

We learn altitude but keep exact waveform design

ML replaces only the outer search; one exact P2 solve preserves feasibility.

**1**

altitude prediction

1

exact P2 solve

**PSD + fixed
diagonal**

feasibility retained by construction

Reproducible data supports two levels of evaluation

1,000

simulated scenarios

8

geometric + control features

800 / 200

train / held-out test

5

independent test splits

h

Input representation

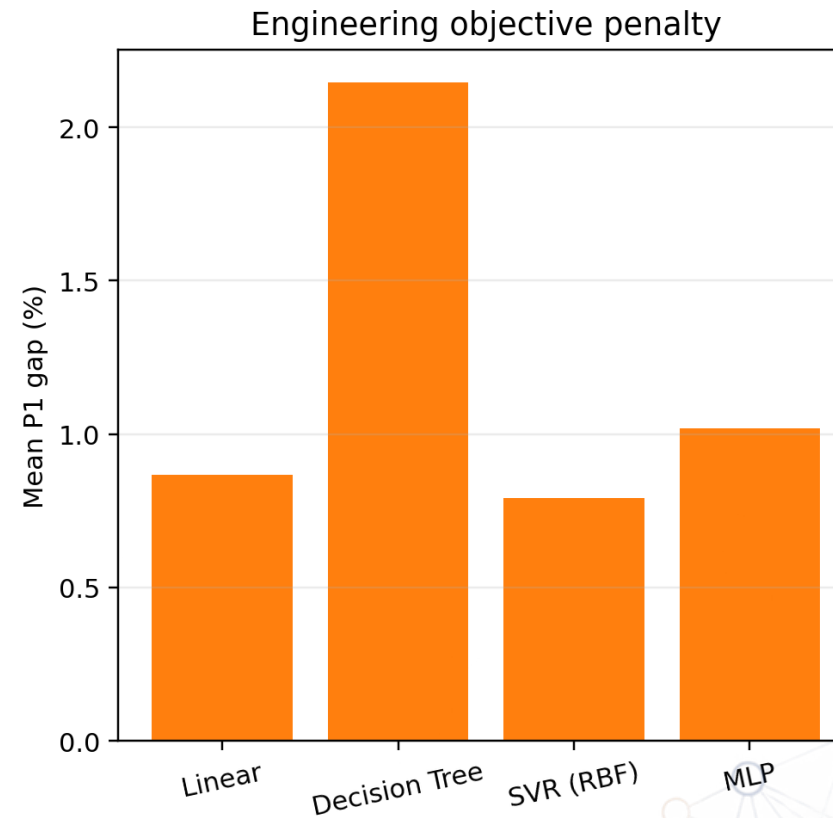
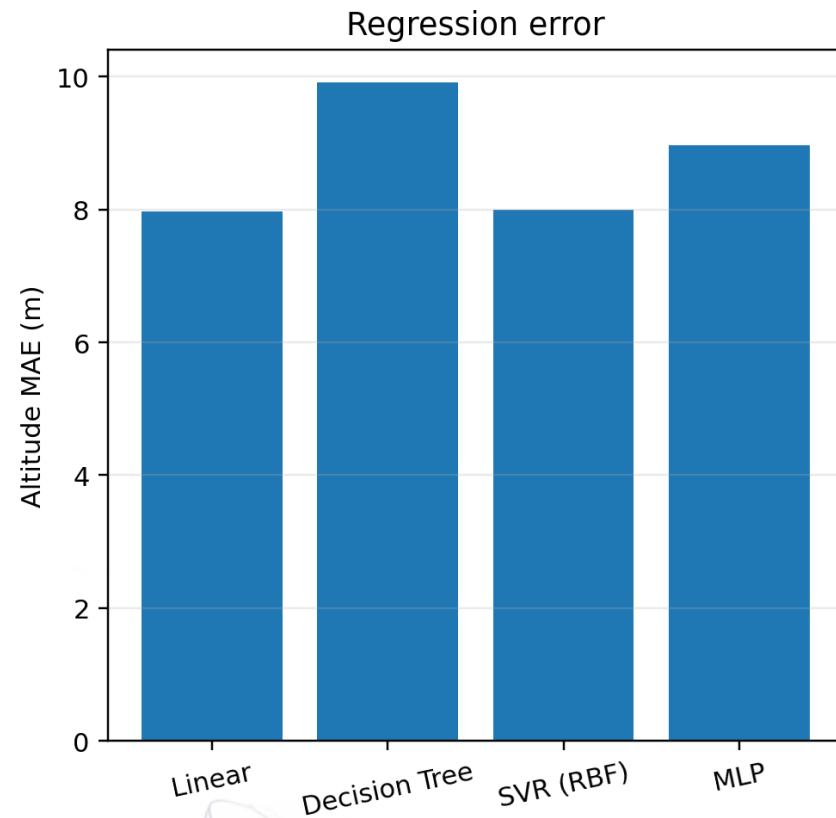
- ρ , target angles, and two EHD positions
- EHDs sorted by distance while preserving each (r, ϕ) pair

Evaluation protocol

- Regression: MAE, RMSE, and R^2 on held-out data
- Engineering: P1 objective gap after one exact P2 solve

Hyperparameters were selected by cross-validation on the training partition only; the test set remained untouched until final evaluation.

Altitude error alone does not select the best surrogate



Selected model

RBF-SVR

7.99 m

seed-42 test MAE

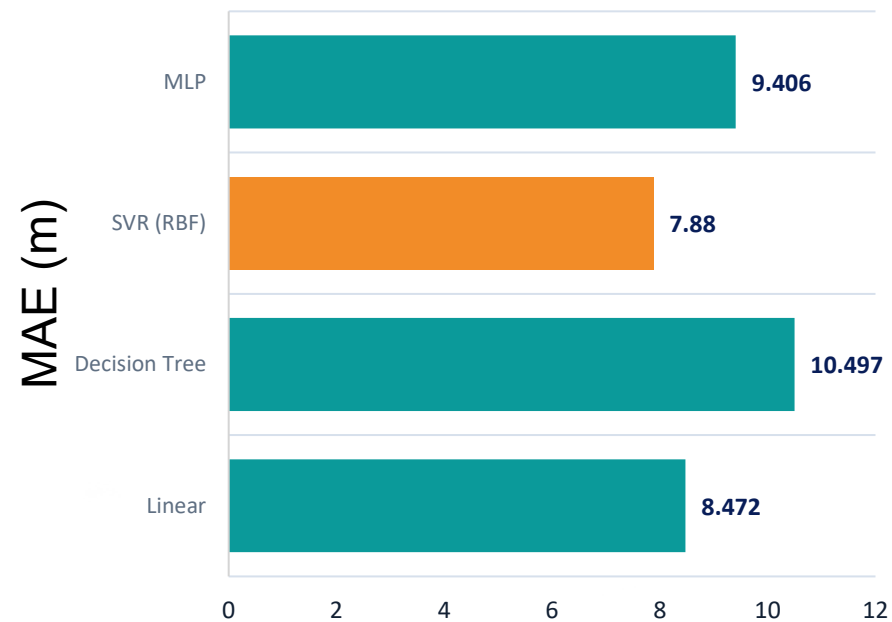
0.793%

mean P1 gap on 200 cases

Key point: a few metres of altitude error can have little objective cost when P1 is locally flat.

SVR is stable across splits; distance dominates

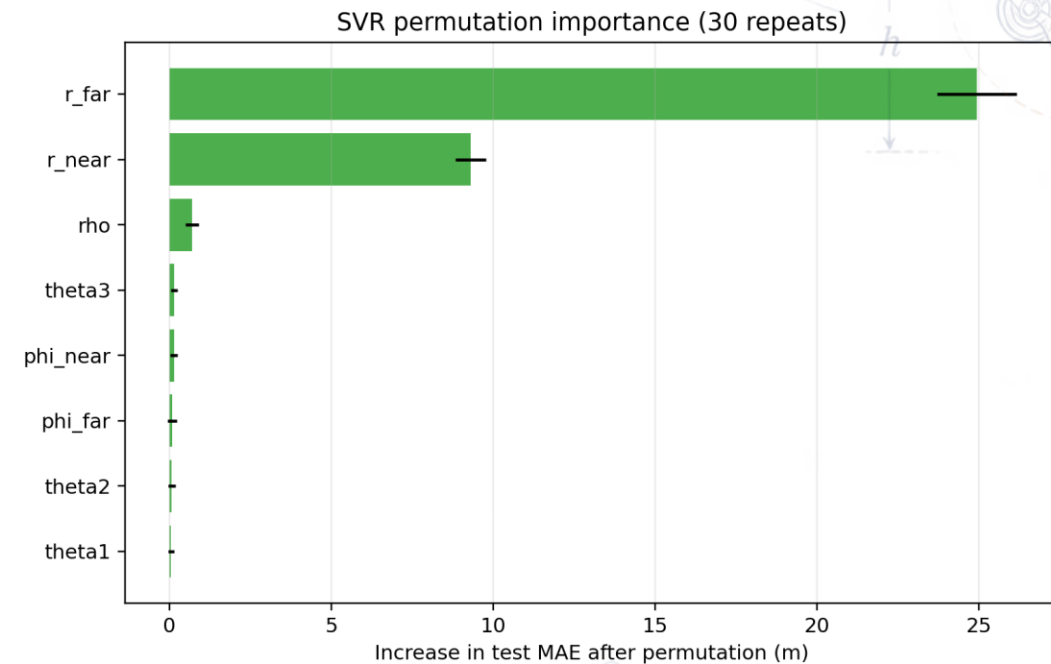
Mean MAE across five held-out splits



7.88 ± 0.67 m

SVR mean $\pm$ standard deviation

Permutation importance of the final SVR

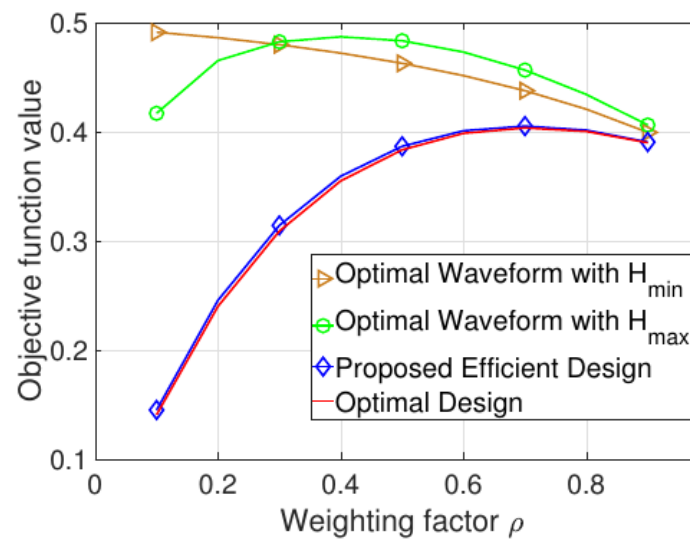


Horizontal distance dominates; angular features contribute much less in this sampled scenario space.

We reproduce Kang before adding the ML alternative

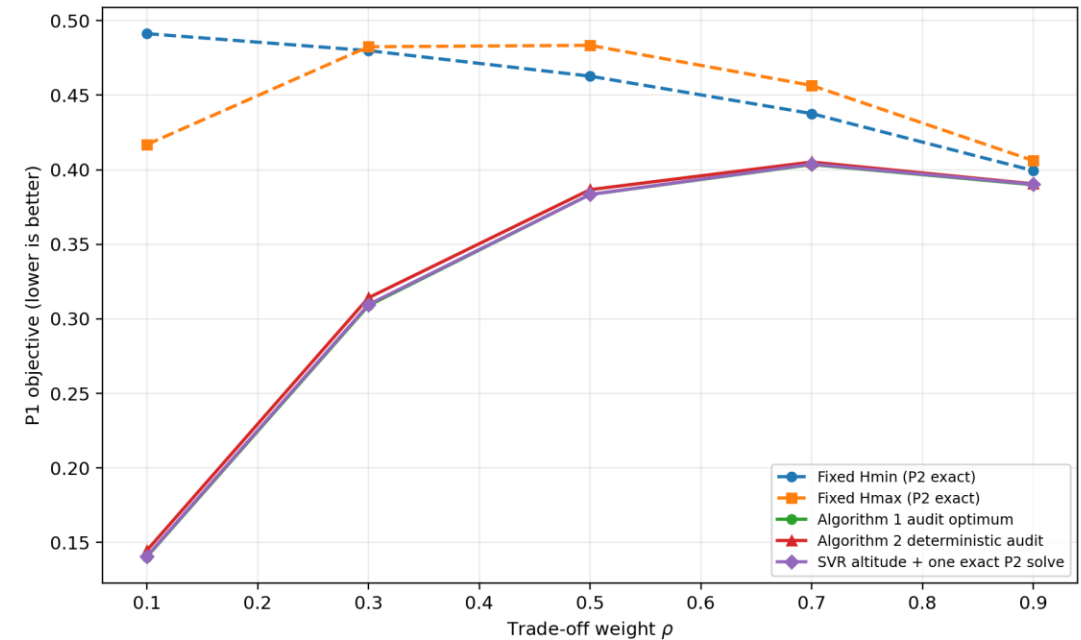
Scenario 1 · target angles $[-40^\circ, 0^\circ, 40^\circ]$ · EHD angles $[-20^\circ, 20^\circ]$ · 0.1° grid

Kang et al. — Figure 3(a)



(a) Objective function value.

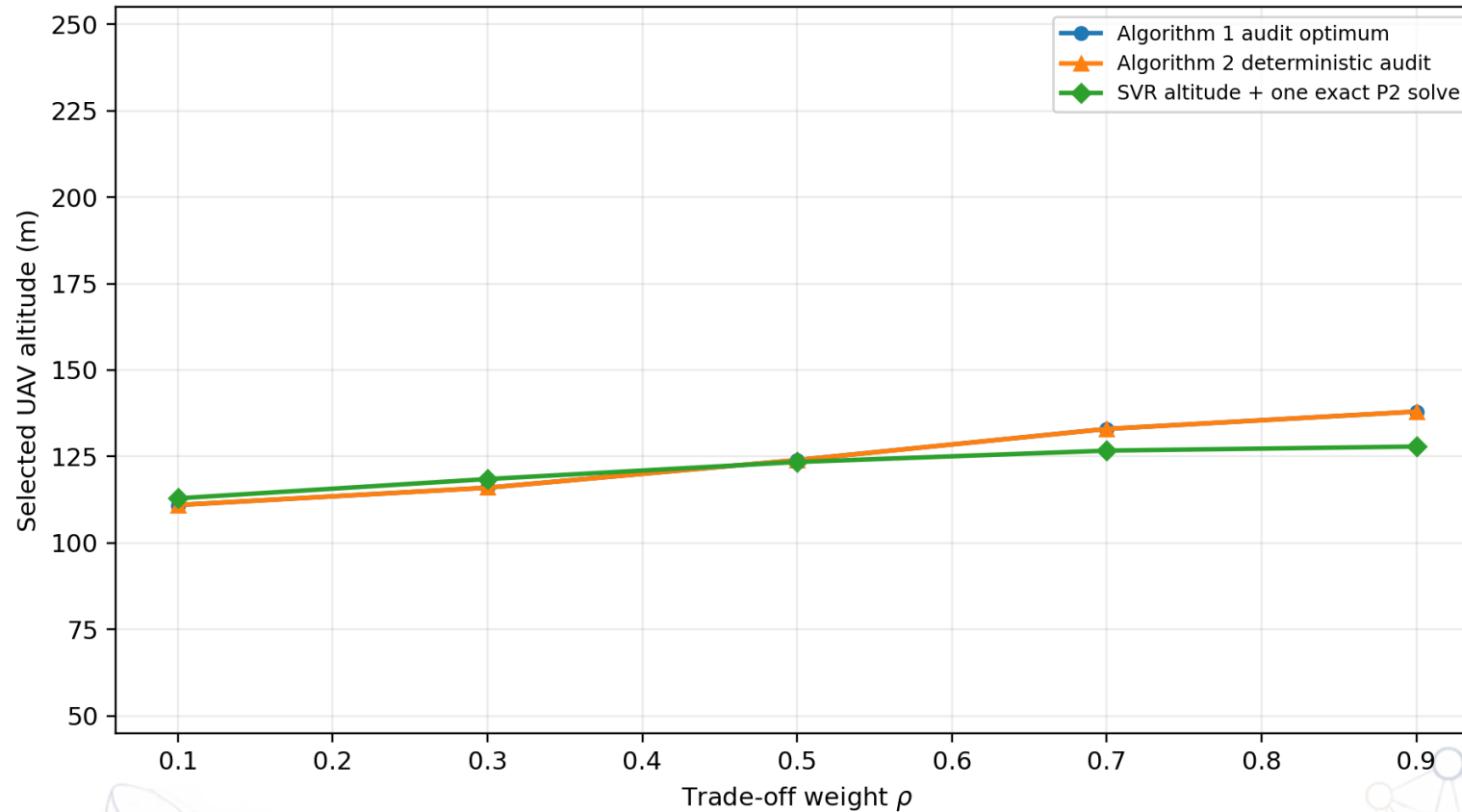
Our verified extension



Across $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, SVR + one exact P2 solve stays within 0.31% of the audit reference.

Left: Kang et al. (2024), Fig. 3(a) · Right: benchmark_gap1.py → gap1_results(1).json

SVR follows the reference altitude trend



Reference altitude

111 → 136 m

as ρ increases

SVR altitude

113 → 128 m

same direction, flatter response

The engineering result remains strong because the optimized objective is relatively flat around the best altitude.

Audits confirm numerical quality and physical feasibility

0.025%

median fast-vs-exact objective error

0.063%

mean label objective gap on 12 exact audits

0

maximum fixed-diagonal violation

Fast waveform solver

- FISTA/Dykstra used for bulk candidate evaluation
- Exact CVXPY P2 retained for audits and final inference

Feasibility checks

- minimum eigenvalue remained non-negative
- per-antenna power diagonal matched exactly

The ML model proposes h ; the optimization layer guarantees a valid waveform X .

Hybrid ML reduces search while preserving constraints

01

Learn the altitude

RBF-SVR replaces the repeated outer search over h .

02

Keep the waveform optimization

One exact P2 solve retains PSD and fixed-power constraints.

03

Evaluate the engineering objective

The selected SVR achieves a 0.793% mean gap on 200 held-out cases and $\leq 0.31\%$ in the Kang Scenario 1 sweep.

73 $\rightarrow$ 1

paper Algorithm 1 PSO budget
 $\rightarrow$ one exact P2 solve at inference

Scope: 73 is the paper PSO budget; the hybrid keeps exact P2 feasibility but does not claim continuous global optimality.

Selected references

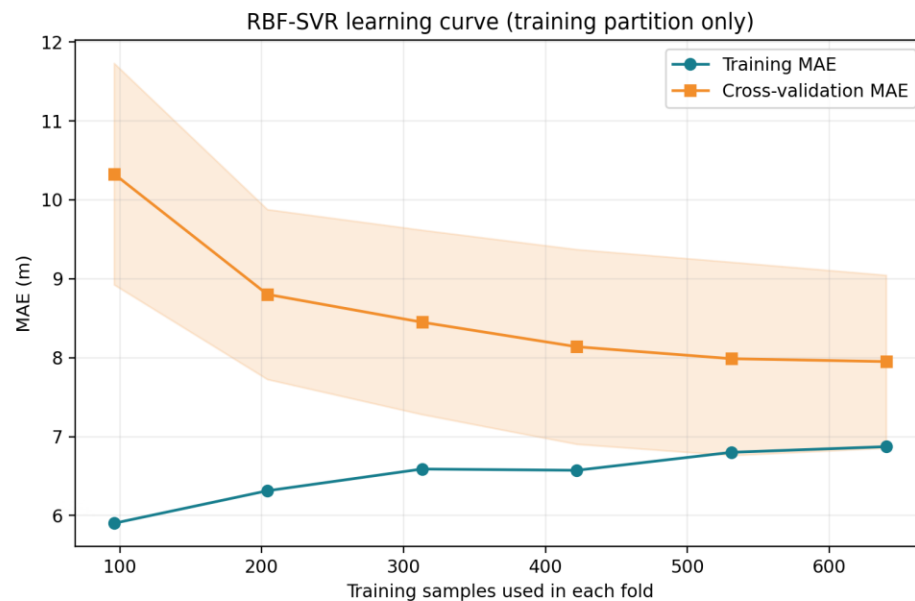
- [1] J. Kang et al., "Joint Design of Transmit Waveform and Altitude for UAV-Enabled Integrated Sensing and Wireless Power Transfer Systems," *Electronics*, vol. 13, no. 21, 4237, 2024.
- [2] H. Sun et al., "Learning to Optimize: Training Deep Neural Networks for Wireless Resource Management," *IEEE Transactions on Signal Processing*, vol. 66, no. 20, pp. 5438–5453, 2018.
- [3] S. Diamond and S. Boyd, "CVXPY: A Python-Embedded Modeling Language for Convex Optimization," *Journal of Machine Learning Research*, vol. 17, no. 83, pp. 1–5, 2016.

- [4] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

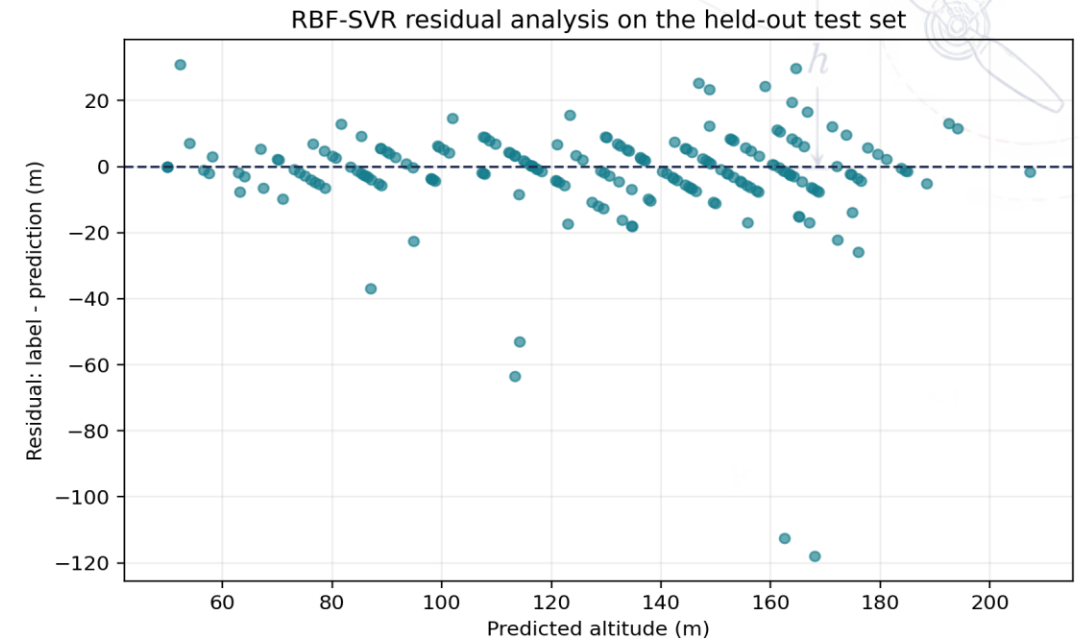
SVR diagnostics separate data sufficiency from outliers

Locked RBF-SVR · 800/200 split · seed 42 · training-only cross-validation

Learning curve



Held-out residuals



CV MAE falls 10.33 → 7.95 m; mean residual is -1.97 m. Engineering impact is evaluated separately with P1.

Source: render_svr_diagnostics.py → svr_diagnostics.json